

and solve the system of equations we get from $\text{grad } \mathcal{L} = \vec{0}$:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial x} &= 20x^{-3/5}y^{1/5}z^{1/5} - 80\lambda = 0, \\ \frac{\partial \mathcal{L}}{\partial y} &= 10x^{2/5}y^{-4/5}z^{1/5} - 12\lambda = 0, \\ \frac{\partial \mathcal{L}}{\partial z} &= 10x^{2/5}y^{1/5}z^{-4/5} - 10\lambda = 0, \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= -(80x + 12y + 10z - 24,000) = 0.\end{aligned}$$

Simplifying this system gives

$$\begin{aligned}\lambda &= \frac{1}{4}x^{-3/5}y^{1/5}z^{1/5}, \\ \lambda &= \frac{5}{6}x^{2/5}y^{-4/5}z^{1/5}, \\ \lambda &= x^{2/5}y^{1/5}z^{-4/5},\end{aligned}$$

$$80x + 12y + 10z = 24,000.$$

Eliminating z from the first two equations gives $x = 0.3y$. Eliminating x from the second and third equations gives $z = 1.2y$. Substituting for x and z into $80x + 12y + 10z = 24,000$ gives

$$80(0.3y) + 12y + 10(1.2y) = 24,000,$$

so $y = 500$. Then $x = 150$ and $z = 600$, and $f(150, 500, 600) = 4,622$ units.

The graph of the constraint, $80x + 12y + 10z = 24,000$, is a plane. Since the inputs x, y, z must be nonnegative, the graph is a triangle in the first octant, with edges on the coordinate planes. On the boundary of the triangle, one (or more) of the variables x, y, z is zero, so the function f is zero. Thus production is maximized within the budget using $x = 150, y = 500$, and $z = 600$.

Exercises and Problems for Section 15.3

Exercises

In Exercises 1–17, use Lagrange multipliers to find the maximum and minimum values of f subject to the given constraint, if such values exist.

- $f(x, y) = x + y, \quad x^2 + y^2 = 1$
- $f(x, y) = x + 3y + 2, \quad x^2 + y^2 = 10$
- $f(x, y) = (x - 1)^2 + (y + 2)^2, \quad x^2 + y^2 = 5$
- $f(x, y) = x^3 + y, \quad 3x^2 + y^2 = 4$
- $f(x, y) = 3x - 2y, \quad x^2 + 2y^2 = 44$
- $f(x, y) = 2xy, \quad 5x + 4y = 100$
- $f(x_1, x_2) = x_1^2 + x_2^2, \quad x_1 + x_2 = 1$
- $f(x, y) = x^2 + y, \quad x^2 - y^2 = 1$
- $f(x, y, z) = x + 3y + 5z, \quad x^2 + y^2 + z^2 = 1$
- $f(x, y, z) = x^2 - y^2 - 2z, \quad x^2 + y^2 = z$
- $f(x, y, z) = xyz, \quad x^2 + y^2 + 4z^2 = 12$
- $f(x, y) = x^2 + 2y^2, \quad x^2 + y^2 \leq 4$
- $f(x, y) = x + 3y, \quad x^2 + y^2 \leq 2$
- $f(x, y) = xy, \quad x^2 + 2y^2 \leq 1$
- $f(x, y) = x^3 + y, \quad x + y \geq 1$
- $f(x, y) = (x + 3)^2 + (y - 3)^2, \quad x^2 + y^2 \leq 2$
- $f(x, y) = x^2y + 3y^2 - y, \quad x^2 + y^2 \leq 10$
- Decide whether each point appears to be a maximum, minimum, or neither for the function f constrained by the loop in Figure 15.30.
 - P
 - Q
 - R
 - S

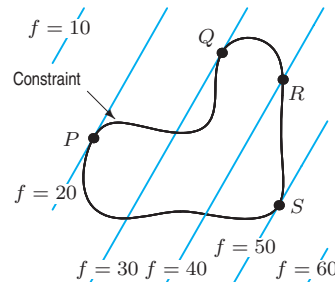


Figure 15.30

Problems

19. Find the maximum value of $f(x, y) = x + y - (x - y)^2$ on the triangular region $x \geq 0, y \geq 0, x + y \leq 1$.
20. (a) Draw contours of $f(x, y) = 2x + y$ for $z = -7, -5, -3, -1, 1, 3, 5, 7$.
 (b) On the same axes, graph the constraint $x^2 + y^2 = 5$.
 (c) Use the graph to approximate the points at which f has a maximum or a minimum value subject to the constraint $x^2 + y^2 = 5$.
 (d) Use Lagrange multipliers to find the maximum and minimum values of $f(x, y) = 2x + y$ subject to $x^2 + y^2 = 5$.
21. Let $f(x, y) = x^\alpha y^{1-\alpha}$ for $0 < \alpha < 1$. Find the value of α such that the maximum value of f on the line $2x + 3y = 6$ occurs at $(1.5, 1)$.
22. Figure 15.31 shows contours of f . Does f have a maximum value subject to the constraint $g(x, y) = c$ for $x \geq 0, y \geq 0$? If so, approximately where is it and what is its value? Does f have a minimum value subject to the constraint? If so, approximately where and what?
24. Figure 15.32 shows ∇f for a function $f(x, y)$ and two curves $g(x, y) = 1$ and $g(x, y) = 2$. Mark the following:
- The point(s) A where f has a local maximum.
 - The point(s) B where f has a saddle point.
 - The point C where f has a maximum on $g = 1$.
 - The point D where f has a minimum on $g = 1$.
 - If you used Lagrange multipliers to find C , what would the sign of λ be? Why?

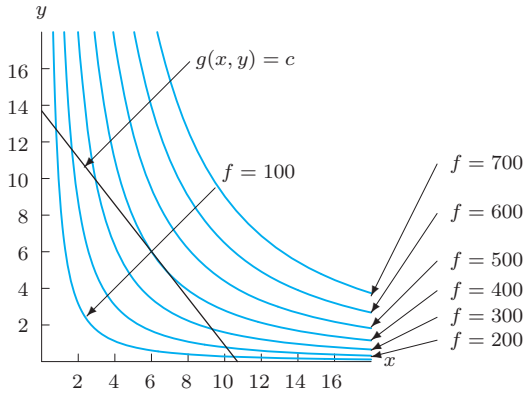


Figure 15.31

23. Each person tries to balance his or her time between leisure and work. The tradeoff is that as you work less your income falls. Therefore each person has *indifference curves* which connect the number of hours of leisure, l , and income, s . If, for example, you are indifferent between 0 hours of leisure and an income of \$1125 a week on the one hand, and 10 hours of leisure and an income of \$750 a week on the other hand, then the points $l = 0, s = 1125$, and $l = 10, s = 750$ both lie on the same indifference curve. Table 15.3 gives information on three indifference curves, I, II, and III.

Table 15.3

Weekly income			Weekly leisure hours		
I	II	III	I	II	III
1125	1250	1375	0	20	40
750	875	1000	10	30	50
500	625	750	20	40	60
375	500	625	30	50	70
250	375	500	50	70	90

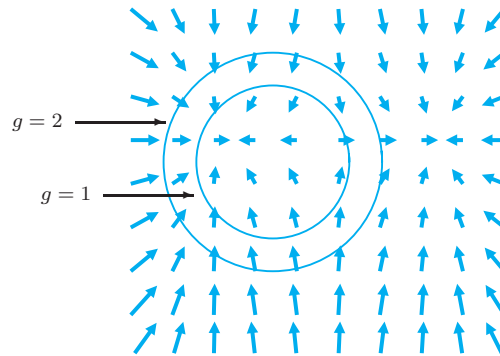
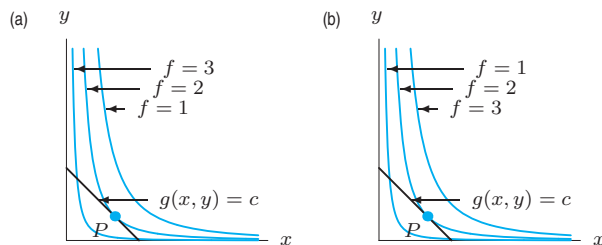


Figure 15.32

25. The point P is a maximum or minimum of the function f subject to the constraint $g(x, y) = x + y = c$, with $x, y \geq 0$. For the graphs (a) and (b), does P give a maximum or a minimum of f ? What is the sign of λ ? If P gives a maximum, where does the minimum of f occur? If P gives a minimum, where does the maximum of f occur?



26. Figure 15.33 shows the optimal point (marked with a dot) in three optimization problems with the same constraint. Arrange the corresponding values of λ in increasing order. (Assume λ is positive.)

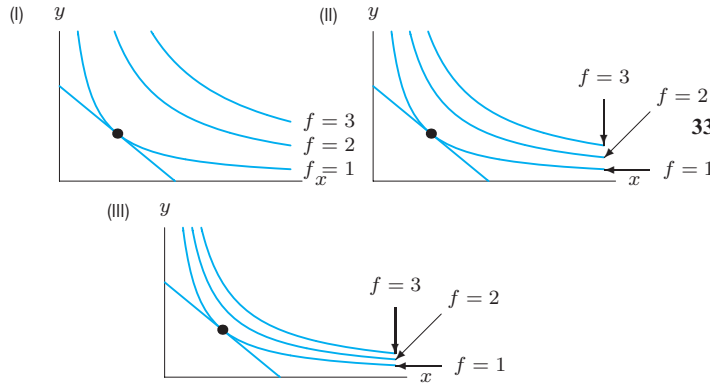


Figure 15.33

27. If the right side of the constraint in Exercise 5 is changed by the small amount Δc , by approximately how much do the maximum and minimum values change?
28. If the right side of the constraint in Exercise 16 is changed by the small amount Δc , by approximately how much do the maximum and minimum values change?
29. The function $P(x, y)$ gives the number of units produced and $C(x, y)$ gives the cost of production.
- A company wishes to maximize production at a fixed cost of \$50,000. What is the objective function f ? What is the constraint equation? What is the meaning of λ in this situation?
 - A company wishes to minimize costs at a fixed production level of 2000 units. What is the objective function f ? What is the constraint equation? What is the meaning of λ in this situation?
30. Design a closed cylindrical container which holds 100 cm^3 and has the minimal possible surface area. What should its dimensions be?
31. A company manufactures x units of one item and y units of another. The total cost in dollars, C , of producing these two items is approximated by the function

$$C = 5x^2 + 2xy + 3y^2 + 800.$$

- If the production quota for the total number of items (both types combined) is 39, find the minimum production cost.
 - Estimate the additional production cost or savings if the production quota is raised to 40 or lowered to 38.
32. An international organization must decide how to spend the \$2000 they have been allotted for famine relief in a remote area. They expect to divide the money between

buying rice at \$5/sack and beans at \$10/sack. The number, P , of people who would be fed if they buy x sacks of rice and y sacks of beans is given by

$$P = x + 2y + \frac{x^2 y^2}{2 \cdot 10^8}.$$

What is the maximum number of people that can be fed, and how should the organization allocate its money?

33. The quantity, q , of a product manufactured depends on the number of workers, W , and the amount of capital invested, K , and is given by

$$q = 6W^{3/4}K^{1/4}.$$

Labor costs are \$10 per worker and capital costs are \$20 per unit, and the budget is \$3000.

- What are the optimum number of workers and the optimum number of units of capital?
 - Show that at the optimum values of W and K , the ratio of the marginal productivity of labor ($\partial q / \partial W$) to the marginal productivity of capital ($\partial q / \partial K$) is the same as the ratio of the cost of a unit of labor to the cost of a unit of capital.
 - Recompute the optimum values of W and K when the budget is increased by one dollar. Check that increasing the budget by \$1 allows the production of λ extra units of the good, where λ is the Lagrange multiplier.
34. A neighborhood health clinic has a budget of \$600,000 per quarter. The director of the clinic wants to allocate the budget to maximize the number of patient visits, V , which is given as a function of the number of doctors, D , and the number of nurses, N , by

$$V = 1000D^{0.6}N^{0.3}.$$

A doctor gets \$40,000 per quarter; nurses get \$10,000 per quarter.

- Set up the director's constrained optimization problem.
 - Describe, in words, the conditions which must be satisfied by $\partial V / \partial D$ and $\partial V / \partial N$ for V to have an optimum value.
 - Solve the problem formulated in part (a).
 - Find the value of the Lagrange multiplier and interpret its meaning in this problem.
 - At the optimum point, what is the marginal cost of a patient visit (that is, the cost of an additional visit)? Will that marginal cost rise or fall with the number of visits? Why?
35. (a) In Problem 33, does the value of λ change if the budget changes from \$3000 to \$4000?
- (b) In Problem 34, does the value of λ change if the budget changes from \$600,000 to \$700,000?
- (c) What condition must a Cobb-Douglas production function, $Q = cK^aL^b$, satisfy to ensure that the

marginal increase of production (that is, the rate of increase of production with budget) is not affected by the size of the budget?

36. The production function $P(K, L)$ gives the number of pairs of skis produced per week at a factory operating with K units of capital and L units of labor. The contour diagram for P is in Figure 15.34; the parallel lines are budget constraints for budgets, B , in dollars.

- (a) On each budget constraint, mark the point that gives the maximum production.
 (b) Complete the table, where the budget, B , is in dollars and the maximum production is the number of pairs of skis to be produced each week.

B	2000	4000	6000	8000	10000
M					

- (c) Estimate the Lagrange multiplier $\lambda = dM/dB$ at a budget of \$6000. Give units for the multiplier.

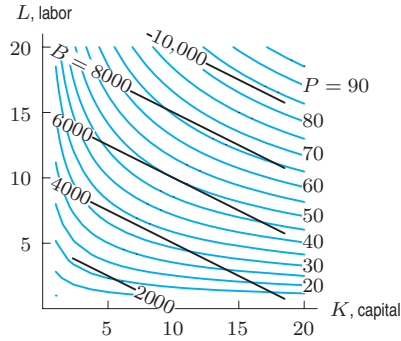


Figure 15.34

37. A doctor wants to schedule visits for two patients who have been operated on for tumors so as to minimize the expected delay in detecting a new tumor. Visits for patients 1 and 2 are scheduled at intervals of x_1 and x_2 weeks, respectively. A total of m visits per week is available for both patients combined.

The recurrence rates for tumors for patients 1 and 2 are judged to be v_1 and v_2 tumors per week, respectively. Thus, $v_1/(v_1 + v_2)$ and $v_2/(v_1 + v_2)$ are the probabilities that patient 1 and patient 2, respectively, will have the next tumor. It is known that the expected delay in detecting a tumor for a patient checked every x weeks is $x/2$. Hence, the expected detection delay for both patients combined is given by¹⁰

$$f(x_1, x_2) = \frac{v_1}{v_1 + v_2} \cdot \frac{x_1}{2} + \frac{v_2}{v_1 + v_2} \cdot \frac{x_2}{2}.$$

Find the values of x_1 and x_2 in terms of v_1 and v_2 that minimize $f(x_1, x_2)$ subject to the fact that m , the number of visits per week, is fixed.

¹⁰Adapted from Daniel Kent, Ross Shachter, *et al.*, Efficient Scheduling of Cystoscopies in Monitoring for Recurrent Bladder Cancer in *Medical Decision Making* (Philadelphia: Hanley and Belfus, 1989).

38. What is the value of the Lagrange multiplier in Problem 37? What are the units of λ ? What is its practical significance to the doctor?

39. Figure 15.35 shows two weightless springs with spring constants k_1 and k_2 attached between a ceiling and floor without tension or compression. A mass m is placed between the springs which settle into equilibrium as in Figure 15.36. The magnitudes f_1 and f_2 of the forces of the springs on the mass minimize the complementary energy

$$\frac{f_1^2}{2k_1} + \frac{f_2^2}{2k_2}$$

subject to the force balance constraint $f_1 + f_2 = mg$.

- (a) Determine f_1 and f_2 by the method of Lagrange multipliers.
 (b) If you are familiar with Hooke's law, find the meaning of λ .

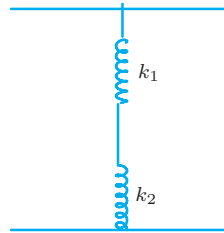


Figure 15.35

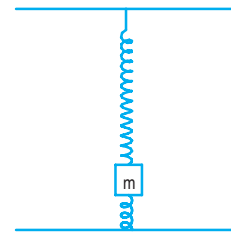
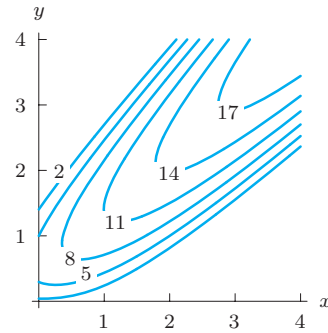
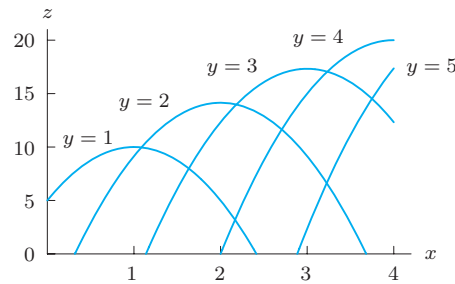


Figure 15.36

40. (a) If $\sum_{i=1}^3 x_i = 1$, find the values of x_1, x_2, x_3 making $\sum_{i=1}^3 x_i^2$ minimum.
 (b) Generalize the result of part (a) to find the minimum value of $\sum_{i=1}^n x_i^2$ subject to $\sum_{i=1}^n x_i = 1$.
41. Let $f(x, y) = ax^2 + bxy + cy^2$. Show that the maximum value of $f(x, y)$ subject to the constraint $x^2 + y^2 = 1$ is equal to λ , the Lagrange multiplier.
42. Find the minimum distance from the point $(1, 2, 10)$ to the paraboloid given by the equation $z = x^2 + y^2$. Give a geometric justification for your answer.
43. A company produces one product from two inputs (for example, capital and labor). Its production function $g(x, y)$ gives the quantity of the product that can be produced with x units of the first input and y units of the second. The cost function (or expenditure function) is the three-variable function $C(p, q, u)$ where p and q are the unit prices of the two inputs. For fixed p, q , and u , the value $C(p, q, u)$ is the minimum of $f(x, y) = px + qy$ subject to the constraint $g(x, y) = u$.
- (a) What is the practical meaning of $C(p, q, u)$?
 (b) Find a formula for $C(p, q, u)$ if $g(x, y) = xy$.

44. A utility function $U(x, y)$ for two items gives the utility (benefit) to a consumer of x units of item 1 and y units of item 2. The indirect utility function is the three-variable function $V(p, q, I)$ where p and q are the unit prices of the two items. For fixed p, q , and I , the value $V(p, q, I)$ is the maximum of $U(x, y)$ subject to the constraint $px + qy = I$.
- What is the practical meaning of $V(p, q, I)$?
 - The Lagrange multiplier λ that arises in the maximization defining V is called the *marginal utility of money*. What is its practical meaning?
 - Find formulas for $V(p, q, I)$ and λ if $U(x, y) = xy$.
45. The function $h(x, y) = x^2 + y^2 - \lambda(2x + 4y - 15)$ has a minimum value $m(\lambda)$ for each value of λ .
- Find $m(\lambda)$.
 - For which value of λ is $m(\lambda)$ the largest and what is that maximum value?
 - Find the minimum value of $f(x, y) = x^2 + y^2$ subject to the constraint $2x + 4y = 15$ using the method of Lagrange multipliers and evaluate λ .
 - Compare your answers to parts (b) and (c).
46. Let f be differentiable and $\text{grad } f(2, 1) = -3\vec{i} + 4\vec{j}$. You want to see if $(2, 1)$ is a candidate for the maximum and minimum values of f subject to a constraint satisfied by the point $(2, 1)$.
- Show $(2, 1)$ is not a candidate if the constraint is $x^2 + y^2 = 5$.
 - Show $(2, 1)$ is a candidate if the constraint is $(x - 5)^2 + (y + 3)^2 = 25$. From a sketch of the contours for f near $(2, 1)$ and the constraint, decide whether $(2, 1)$ is a candidate for a maximum or minimum.
 - Do the same as part (b), but using the constraint $(x + 1)^2 + (y - 5)^2 = 25$.
47. A person's satisfaction from consuming a quantity x_1 of one item and a quantity x_2 of another item is given by
- $$S = u(x_1, x_2) = a \ln x_1 + (1 - a) \ln x_2,$$
- where a is a constant, $0 < a < 1$. The prices of the two items are p_1 and p_2 respectively, and the budget is b .
- Express the maximum satisfaction that can be achieved as a function of p_1, p_2 , and b .
 - Find the amount of money that must be spent to achieve a particular level of satisfaction, c , as a function of p_1, p_2 , and c .
48. This problem illustrates the Envelope Theorem, which relates the maxima of $z = f(x, y)$ subject to the constraint $x = c$ to the contour diagram in Figure 15.37 and the cross-sections in Figure 15.38.
- For each value c , there is a maximum value of $f(x, y)$ with $x = c$. On Figure 15.37, sketch the curve that goes through the points where the maxima are achieved.
 - On Figure 15.38, sketch the curve going through the points corresponding to the same maximum values in part (a). This curve is called the envelope of the cross-sections.
 - Show that the Lagrange multiplier λ for this constrained optimization problem is the slope of the envelope curve in part (b).


 Figure 15.37: Contour diagram of f

 Figure 15.38: Cross-sections of f

Strengthen Your Understanding

In Problems 49–50, explain what is wrong with the statement.

49. The function $f(x, y) = xy$ has a maximum of 2 on the constraint $x + y = 2$.
50. If the level curves of $f(x, y)$ and the level curves of $g(x, y)$ are not tangent at any point on the constraint $g(x, y) = c, x \geq 0, y \geq 0$, then f has no maximum on the constraint.

In Problems 51–55, give an example of:

51. A function $f(x, y)$ whose maximum subject to the constraint $x^2 + y^2 = 5$ is at $(3, 4)$.
52. A function $f(x, y)$ to be optimized with constraint $x^2 + 2y^2 \leq 1$ such that the minimum value does not change when the constraint is changed to $x^2 + y^2 \leq 1 + c$ for $c > 0$.